

2026 Science Olympiad @ Berkeley Invitational

Heredity B Test



Exploring the World of Science

School Name: _____

Team Name: _____

Team Number: _____

Student Names: _____

Instructions:

- You have 50 minutes to complete this exam.
- You can write in and separate the pages of this booklet
- Each team is permitted one, double-sided, letter-sized cheat sheet and two calculators per the rules.
- Tiebreakers are indicated and will be broken based on the number of points earned on each tiebreaker question in order.
- Good luck and have fun!

Genetic Code

Second Letter

		U	C	A	G		
First Letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG } Stop	UGU } Cys UGC } UGA } Stop UGG } Trp	U	Third Letter
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	C	
	A	AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	A	
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } Asp GGC } GGA } Glu GGG }	G	

Use the following gene information to answer questions

For Gene A, A - 2 eyes and a - 3 eyes

For Gene B, B - 2 arms and b - 3 arms

For Gene C, C - 2 legs and c - 3 legs

2. (3) You cross two individuals with the genotypes Aabb x AABb. What are the genotypic and phenotypic ratios? (Order from largest to smallest numerically. Ex: 3:2:1 not 1:2:3 or 2:1:3)

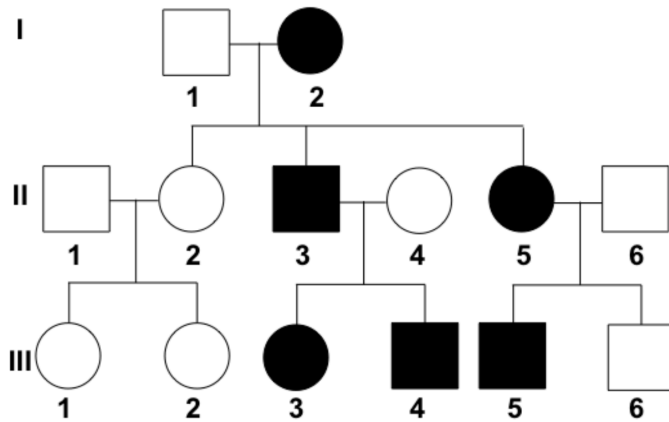
3. (1 each) Predict the probability of the following genotypes and phenotypes of the offspring from aaBbCCDD x AaBbCcdd. (Tiebreaker 2)

- (a) AABBCC _____
- (b) AaBBCCc _____
- (c) aaBbCC _____
- (d) 2 eyes, 2 arms, 2 legs _____
- (e) 3 arms _____
- (f) 3 eyes, 3 arms, 3 legs _____

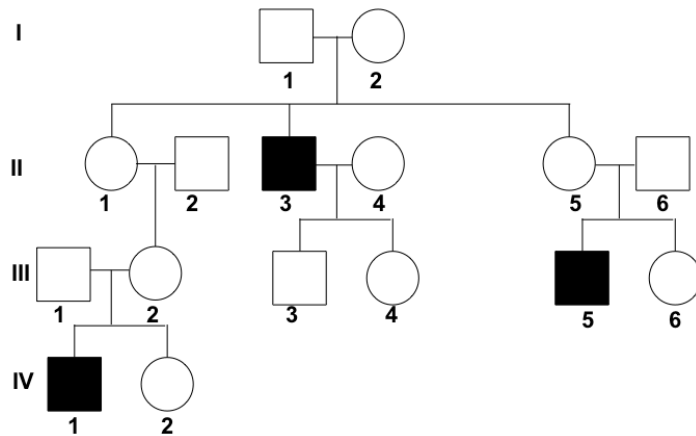
4. (2) You cross two individuals that are both heterozygous for genes A through Z, which is 26 genes in total. What is the probability that an offspring is also heterozygous for genes A-Z? Give your answer as an expression. (Don't calculate anything).

5. (2) A male with blood type AB is crossed with a female of the genotype $I^B i$. What is the probability that the offspring is type B?

8. (2 each) What modes of inheritance are displayed in the pedigrees?



a. _____



b. _____

Mitosis and Meiosis

9. (1) What is the order of the stages in mitosis?

10. (1) What are the stages of the cell cycle in order?

11. (2) A cell ($2n = 8$) undergoes meiosis 1 and meiosis 2. How many chromosomes do the daughter cells have after meiosis 2?

12. (2) What phenomenon is responsible for the above daughter cells being genetically different?

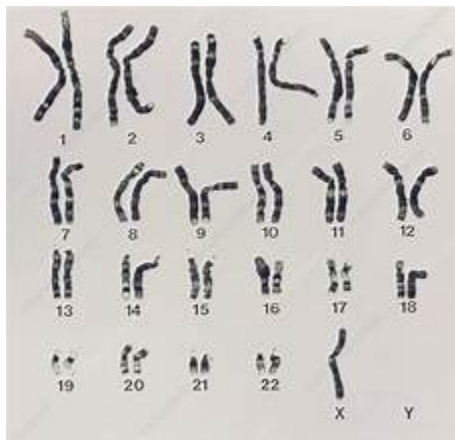
13. (1 each) Give the structures to the following descriptions in mitosis and meiosis. (Tiebreaker 4)

- a. DNA sequence that holds sister chromatids together _____
- b. Attaches chromosomes to microtubules _____
- c. Pulls chromosomes apart in anaphase _____
- d. Microtubule-organizing centers _____
- e. Equivalent of cleavage furrow in plant cells _____

14. (2) What is the term for uneven splitting of chromosomes during anaphase?

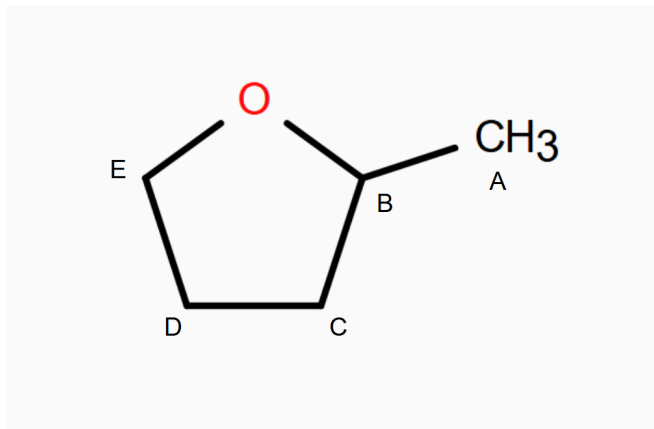
15. (1) Which of Mendel's laws does nondisjunction violate?

16. (1) What is the genetic disease depicted in the karyotype below?



17. (1) What are the implications of an absent **Y** chromosome in the individual?

Molecular Biology of DNA



18. (3) The pentose sugar of a nucleotide is shown. What three structures are attached to the pentose sugar which would fully comprise a nucleotide? (You can ignore the letters)

19. (1) How are deoxyriboses in DNA different from riboses in RNA?

20. (2) Given the DNA sequence 5'- GATACAGATACA - 3'. What is the complementary DNA strand?

5' - _____ - 3'

21. (1 each) Match the enzyme with its function in DNA replication. (Tiebreaker 3)

1. Helicase
2. Single-stranded binding proteins
3. Primase
4. DNA Polymerase
5. Ligase

6. Topoisomerase

- a. ___ Binds to separated single DNA strands
- b. ___ Adds nucleotides to synthesize new DNA strands
- c. ___ Joins Okazaki fragments by forming phosphodiester bonds
- d. ___ Unwinds DNA double helix
- e. ___ Synthesizes RNA primers for initiation of DNA synthesis
- f. ___ Relieves tension generated from DNA supercoiling

22. (3) Rank the following in order from least compact to most compact. Just use the numbers.

- 1. Euchromatin
- 2. Chromosomes
- 3. DNA double helix
- 4. Heterochromatin

Prokaryotic and eukaryotic gene expression and regulation

24. (4) If a DNA double helix has 32% adenine, what percent of thymine, guanine, cytosine, and uracil does it contain?

Fill in the following blanks

25. (5) The enzyme responsible for transcription is _____, and this process takes place in the _____. The 3 stages of transcription are _____, _____, _____.

26. (5) Translation occurs in the _____ (location) where it is carried out by _____ (organelle). In the _____ site, the tRNA enters, and the polypeptide is transferred from the tRNA already in the _____ site. Both tRNAs shift one site over, and the tRNA that previously had the polypeptide exits through the _____ site.

27. (1) Name an enzyme that violates the Central Dogma.

28. (2) What is the name of the phenomenon that allows for multiple codons to code for the same amino acid?

30. (3) Given the template strand below, what would be the resulting amino acid sequence?

TACGCATAACTGGTGTACATT

- a. (1) What is the mRNA transcribed from the template strand
- b. (2) What is the amino acid sequence translated from the mRNA

Technology and techniques

31. (3) What are the 3 stages of PCR, and what temperatures are they performed at?

32. (2) What special DNA polymerase is used in PCR?

33. (1) DNA polymerase can not initiate DNA synthesis by itself. What does polymerase add onto to synthesize DNA in PCR?

34. (2) How many DNA double helices are present after one DNA molecule undergoes 5 PCR rounds?

35. (1) What did one DNA double helix say to the other DNA double helix? (this is meant to be a joke)

36. (1) What are you interested in competing in Heredity B? (wrong answers only)